Engineering - Undergraduate Level Assessment 13 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. A liquid compressed in a cylinder has a volume of 1 liter (l) (1000 cm^3) at 1 MN/m^2 and a volume of 995 cm^3 at 2 MN/m^2. What is the bulk modulus of elasticity?
   (a) 400 Mpa
   (b) 600 Mpa
   (c) 750 Mpa
   (d) 150 Mpa
   (e) 500 Mpa

2. Find the general solution of Bessel's equation of order one.
   (a) y(x) = c_1 * sin(x) + c_2 * cos(x)
   (b) y(x) = c_1 * y_1$(x)^2$ + c_2 * y_2$(x)^2$
   (c) y(x) = c_1 y_1(x) + c_2 y_2(x)
   (d) y(x) = c_1 * x * y_1(x) + c_2 / (x * y_2(x))
   (e) y(x) = c_1 * y_1(x) / (c_2 * y_2(x))

3. Determine the mean-square quantization error during the quantization of a signal. Assume that the error (equally likely) lies in the range - S/2 to + S/2, where S is the step size.
   (a) (S^2 / 3)
   (b) (S^2 / 12)
   (c) (S^2 / 24)
   (d) (S^2 / 8)
   (e) (S^2 / 18)

4. Calculate the Laplace transform $oft^n u(t)$.
   (a) L{$t^n u(t)$} = [n! / ($s^($n+3))]
   (b) L{$t^n u(t)$} = [(n+1)! / ($s^($n+1))]
   (c) L{$t^n u(t)$} = [n! / (s^n)]
   (d) L{$t^n u(t)$} = [n! / (s^n+ 2)]
   (e) L{$t^n u(t)$} = [n! / (s^n+ 1)]

5. Calculate the change in the stored energy of a parallel plate capacitoras a result of inserting a dielectric slab.

(a) 0.30
(b) 0.92
(c) 0.99
(d) 1.05
(e) 0.65

True / False (10 marks)

Answer True or False

6. True or False: In the force-current analogy, the electrical analogous quantity for a dash-pot is admittance.
7. True or False: The electric field in the gap between two infinite parallel plates is double the field produced by one plate.
8. True or False: The Fourier series f(t) can be equivalently represented as f(t) = 2 + (cos(kt) + k sin(kt)) / (1 + k^2).
9. True or False: The 8085 microprocessor has 22 output pins.
10. True or False: Sensitivity refers to the smallest change in measured variable that an instrument will respond to, similar to resolution.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the loss of head in a 1/4 inch diameter, 16 ft long pipe when water flows at half the critical velocity. Discuss the factors influencing this calculation and the implications for fluid dynamics.
12. A vertical cylinder with a diameter of 300 mm contains water at $70^a ndhasafrictionlesspiston.Analyz$

End of Paper